

SUMMARY Data Engineer with 2+ years of experience building scalable data architectures and production-grade ML solutions. Expert in high-performance data engineering pipelines, AI/ML development, and Large Language Model (LLM) integration. Proficient in streamlining processes through autonomous workflow automation with n8n.

- TECHNICAL SKILLS**
- **Data Engineering:** Apache Airflow, AWS, Azure, Kafka, Spark, PostgreSQL
 - **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, YOLOv8, OpenCV, Keras, Pandas, NumPy
 - **Cloud & DevOps:** AWS (S3, EC2, Lambda, IAM), Docker, Kubernetes, CI/CD (GitHub Actions), Linux, Terraform
 - **Languages:** Python, SQL, JavaScript(React , Next, ReactNative)
 - **Concepts:** ETL/ELT Pipelines, Data Warehousing, Object Detection, REST APIs, Agile, n8n, Langchain

- EXPERIENCE**
- Georgia State University**
Graduate Teaching Assistant

Aug 2025 - Present

- Served as a first point of contact for 50+ students, practicing active listening and providing resource referrals to ensure academic and personal well-being.
 - Collaborated with faculty to modernize course curriculum by integrating real-world datasets, ensuring students gained practical experience with industry-standard tools.
- Merodream**
Data Engineer

Sep 2023 - Aug 2025

- Designed and implemented ETL scripts, CTEs, views, procedures, and Redshift tasks to automate the daily data processing orchestrated via Apache Airflow.
 - Developed an AWS Redshift pipeline to run daily ETL scripts, monitored batch jobs, and resolved issues, ensuring continuous data processing without downtime
 - Optimized SQL queries and implemented table partitioning strategies, cutting cloud compute costs by 25% while handling terabytes of monthly transaction data.
 - Collaborated with ML teams to deploy model training pipelines using Docker and Kubernetes, reducing setup time and improving scalability for model training

- PROJECTS**
- SAM2-LatentDiff**

- Developed a two-stage low-light enhancement system combining frozen SAM2 hierarchical features with a pretrained Stable Diffusionv1.5 U-Net, using cross-attention and LoRA for parameter-efficient latent-space fine-tuning
 - Built an 87K-parameter PixelRefiner to reduce VAE reconstruction artifacts; achieved 18.61 PSNR, 0.7085 SSIM, and 0.2399 LPIPS on the LOL-v2 Real test set.
 - Implemented preprocessing, training, evaluation, visualization, and ablation pipelines; demonstrated 111 ms single-pass inference on an NVIDIA A100.
- Hull Tactical Market Prediction**

- Built a high-performance feature engineering pipeline using Polars and XGBoost (GPU-accelerated), dynamically selecting the top 30 signal features from complex volatility and sentiment interactions.
 - Engineered a robust 6-model stacking ensemble (including CatBoost, LightGBM, and Random Forest) with a RidgeCV meta-learner, optimizing model stability for noisy financial data.
 - Implemented a production-simulation inference server with volatility targeting logic, ensuring consistent model deployment and risk-adjusted scoring in a live-loop environment.
- Wildlife Conservation Image Classification**

- Built an automated species classifier using ResNet50 with Transfer Learning to process camera-trap images from Côte d'Ivoire.
 - Addressed severe class imbalance (90% empty frames) by implementing Focal Loss and oversampling minority classes, boosting recall for rare species by 20%.
 - Developed a “motion-trigger” filtering script to automatically discard empty frames, reducing manual labeling effort by 70%.

EDUCATION

Georgia State University
M.S. Computer Science

Aug 2027

- CERTIFICATIONS**
- AI Open Source Capstone Course: Codepath
 - Supervised Machine Learning: Regression and Classification: Stanford/Coursera
 - Python for Data Science, AI & Development: IBM
 - Diploma in Quantum Computing and Programming (Bronze): Qiskit
 - AWS Academy Graduate – Cloud Foundations
 - Capstone: Retrieving, Processing, and Visualizing Data with Python
 - Python Data Structures: Coursera

Jun 2026

Apr 2024

Oct 2023

Sep 2023

Jul 2023

Jun 2021

Sep 2020

PUBLICATIONS

- DAAL: Density-Aware Adaptive Line Margin Loss for Multi-Modal Deep Metric Learning..*arXiv:2410.05438*